Suphia Zhang

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Education

Cornell University *Ithaca, NY*

B.S. Mechanical Engineering, Minor in Aerospace Engineering *Aug 2025 – May 2029*

- **Awards:** Meinig Family Cornell National Scholar (2025-26), McMullen Scholarship Recipient (2025-26), McMullen Dean's Scholar (2025-26), National First-Generation Recognition Program
- **Relevant Coursework:** Thermodynamics, Statics & Mechanics of Solids, Differential Equations, Heat & Mechanics

Engineering Projects & Research

NHRL Combat Robotics Team (National Havoc Robot League) *Ithaca, NY*

Mechanical Design Lead *Nov 2025 – Present*

- Led mechanical design for a 4-person team building a 3 lb beetleweight combat robot, taking it from hand sketches to a full OnShape CAD model in 6 weeks and meeting the 3 lb weight limit with 1 oz to spare.
- Built prototypes and ran **test drives** to validate drive and weapon subsystems, characterizing battery consumption and tuning weapon control before competition.
- Fabricated and assembled all subsystems (machined parts, carbon fiber layups, cast urethane wheels), with the robot surviving 4 competition matches without structural failure.

Cornell University Physical Intelligence (CUPI) *Ithaca, NY*

Mechanical Co-Lead *Nov 2026 – Present*

Mechanical Design Team Member *Mar 2026 – Nov 2026*

- Designed hexapod leg and tactile foot-sensing subsystems for Symbiote, a bio-inspired multi-robot system pairing an **autonomous UAV quadcopter** with a hexapod ground robot for coordinated terrain surveying.
- Redesigned linkages for all 6 hexapod legs and reinforced the leg carriage in Fusion 360 through iterative CAD and material analysis, increasing range of motion and load capacity for heavier sensor payloads.
- Designed and built a tactile foot sensor (Velostat taxel array with copper and Kapton electrodes in a TPU housing); preparing **load tests with known weights** to characterize its force range and limits.
- Wrote Arduino (C++) and Python **data acquisition** scripts for real-time force visualization, and ran MuJoCo simulations modeling each taxel as a contact to compute center of pressure.

Bio-Inspired Fluid Lab *Ithaca, NY*

Undergraduate Researcher, Advisor: Dr. Sunghwan Jung *Sep 2026 – Present*

- Developing a **flapping-wing aerial-aquatic vehicle** modeled on MIT's FAAV, a $\sim$300 g diving-bird-inspired robot that both swims and flies, within the lab's bird design subgroup.
- Built a simulation assembly of the flapping-wing mechanism in OnShape, defining mates and joint constraints to model wing kinematics and verify range of motion before fabrication.

Work Experience

Double Dragon Restaurant *Pawtucket, RI*

Line Cook / Cashier / Staff *2015 – Aug 2026*

- Worked 60-hour weeks during academic breaks, diagnosing and repairing commercial kitchen equipment including fryers to minimize downtime during peak service.
- Onboarded 3 third-party delivery platforms (DoorDash, Grubhub, Uber Eats) in under a month, handling setup and merchant documentation, generating $\sim$200 online orders per week while managing 30+ orders per shift.

Leadership & Activities

American Society of Mechanical Engineers (ASME), Cornell University *Ithaca, NY*

Corporate Committee Member *Dec 2025 – Present*

- Conduct employer outreach and compile company dossiers to co-organize networking sessions, workshops, and speaker panels for 200+ Cornell ME students.

Skills

Testing & Validation: Prototype testing, data acquisition (Arduino, Python), simulation-based validation (MuJoCo, OnShape), troubleshooting
Fabrication: Carbon fiber layup, machining, 3D printing (PLA, TPU), urethane casting, soldering, assembly
CAD & Design: OnShape, Fusion 360, GD&T, simulation assemblies, design iteration & validation
Programming & Software: Python, MATLAB (Onramp certified), Arduino (C++), MuJoCo, LaTeX

Languages: English (Native), Fuzhounese (Native), Mandarin (Conversational), Spanish (Conversational)